

Exactly — that's a very clear and justifiable way to present it. You can phrase it like this in the blog:

Why not use usual linear regression?

- Linear regression assumes that the **residuals (errors) are normally distributed** and that the model predicts the **mean outcome**.
 - In ultramarathon finish times:
 - The distribution is **heavily left-skewed** (a few very fast pros, majority of slower runners).
 - The target variable has a **long left tail** due to this skew and hard cutoffs (race time limits).
 - Log or other simple transformations do **not normalize** the distribution sufficiently.
 - Using OLS would give **misleading predictions for the tail** (fast runners), and confidence intervals would be unreliable there.
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Why not use AFT survival models?

- AFT can account for censored data (DNF) in a classical survival sense.
 - **But in this context, censoring times are not informative:**
 - A DNF is a DNF, **the actual recorded time doesn't matter**.
 - A runner dropping at hour 20 is no better than one dropping at hour 40 — the “event” of interest is finishing, not surviving longer.
 - Using AFT would imply that “longer survival” (slower finish) is better, which is **the opposite of our race objective**.
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Why quantile regression is ideal

- It directly models **conditional percentiles** of finish time (e.g., 50th, 90th percentile).
 - You can answer questions like: “What UTMB index gives a 90% chance of finishing under 27h?”
 - It **does not assume normality** of residuals.
 - Interpretation is still straightforward: coefficients indicate how features shift the percentile of finish time.
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In short:

“Linear regression fails because the target is heavily skewed. AFT survival modeling is misleading because censoring times don't matter — DNF is DNF. Quantile regression allows us to model specific finish time probabilities accurately and interpretably.”

If you want, I can help you **write this paragraph for your blog in a concise, reader-friendly way**, maybe with a small diagram showing the skew and the percentile approach. It would make it very clear for non-technical readers.

Do you want me to do that?